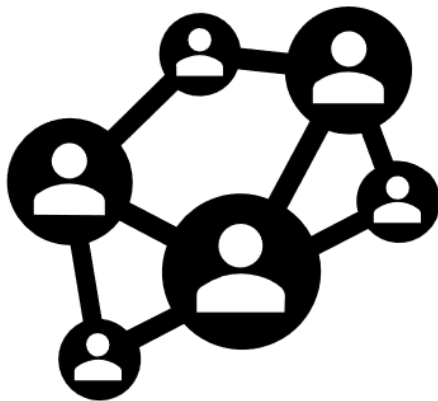


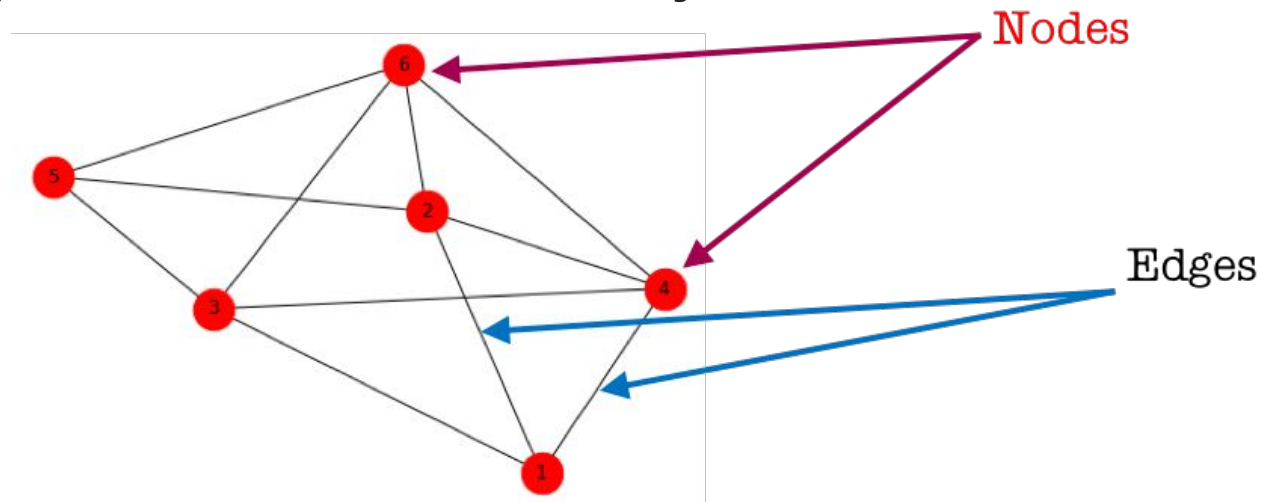


Network Analysis

Graph Data Structure

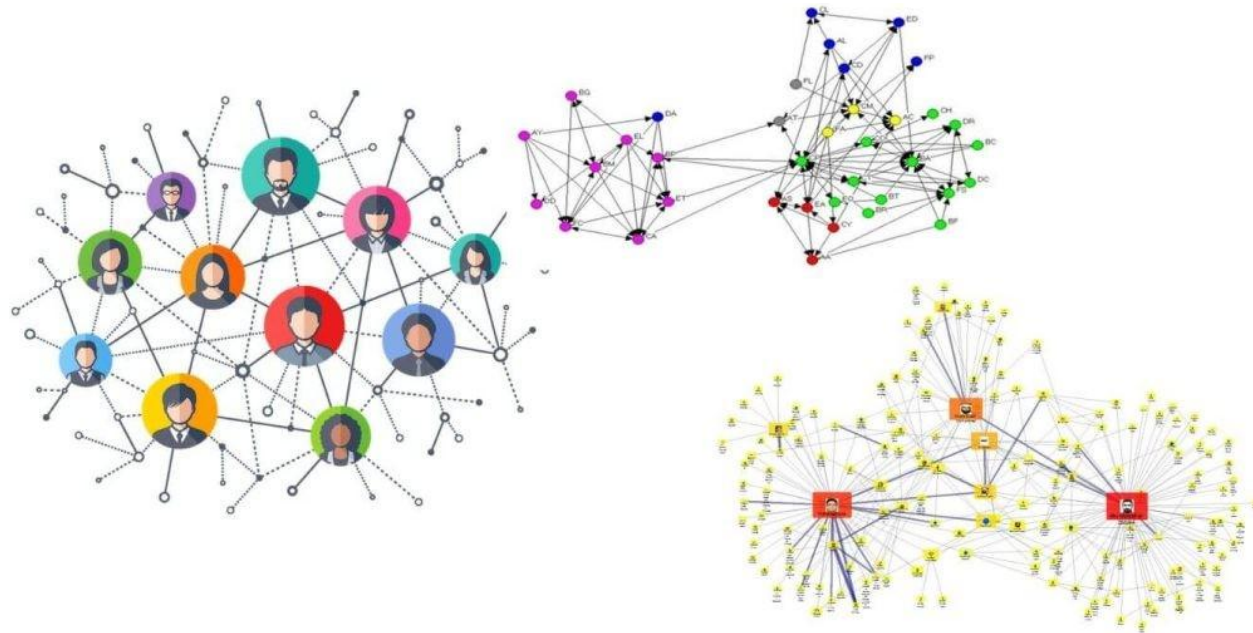
Network Analysis

- ❖ A network refers to a **structure representing** a group of objects/people and **relationships** between them.
- ❖ It is **also known as a graph** in mathematics. A network structure consists of nodes and edges. Here, nodes represent objects we are going to analyze while edges represent the relationships between those objects.



Applications of Network Analysis

- ❖ If you are using any **social media applications**, you may have experienced the friend or follower suggestions functions.
- ❖ Studying a social relationship between Facebook users, nodes are target users and edges are relationships such as friendships between users or group memberships.

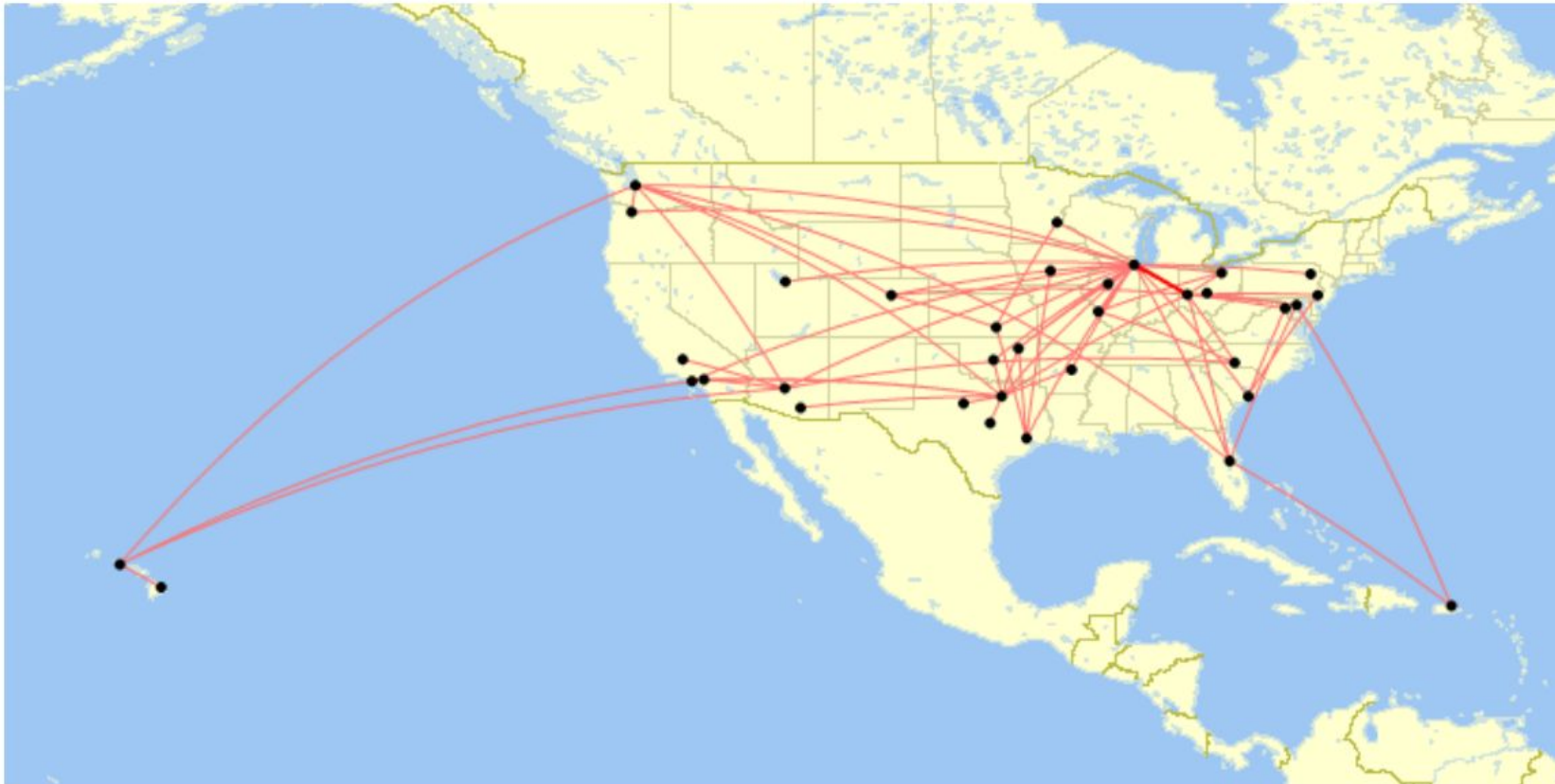


Applications of Network Analysis

- ❖ Helps in understanding the **structure of a relationship** in social networks, a structure or process of change in natural phenomenon, or even the analysis of biological systems of organisms.
- ❖ Specifically:
 - Identifying the most influent person/people in a group
 - Defining characteristics of groups of users
 - Prediction of suitable items for users, etc.

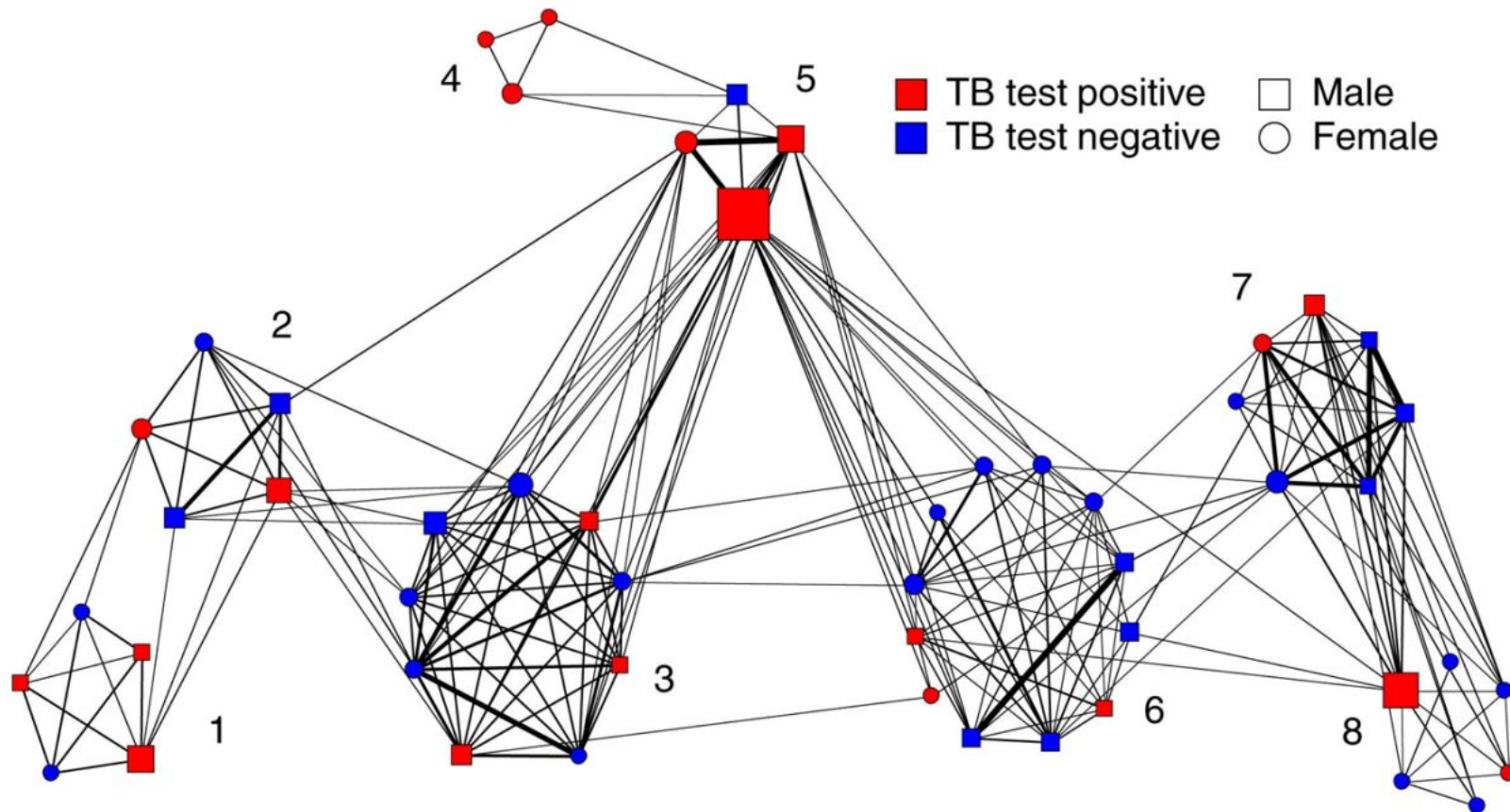
Applications of Network Analysis

Example: Flight Routes



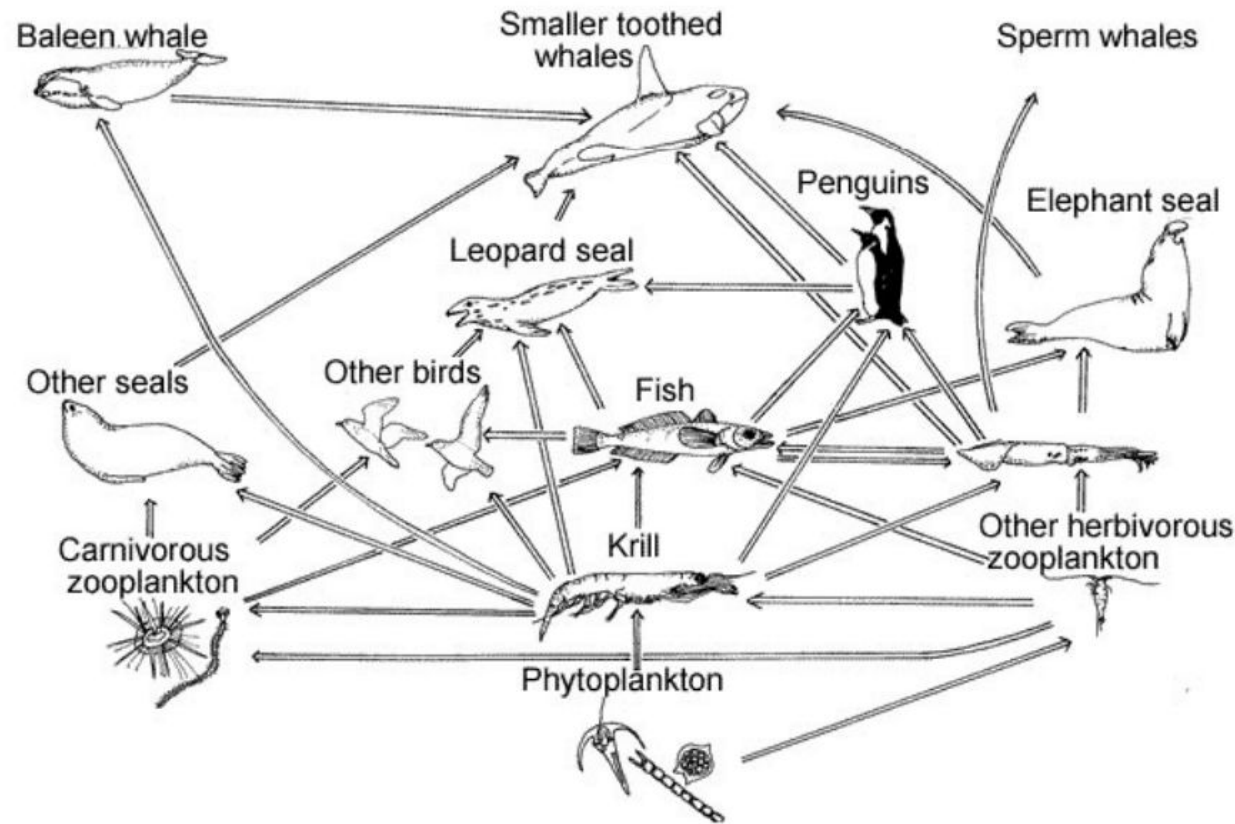
Applications of Network Analysis

Example: Disease Transmission



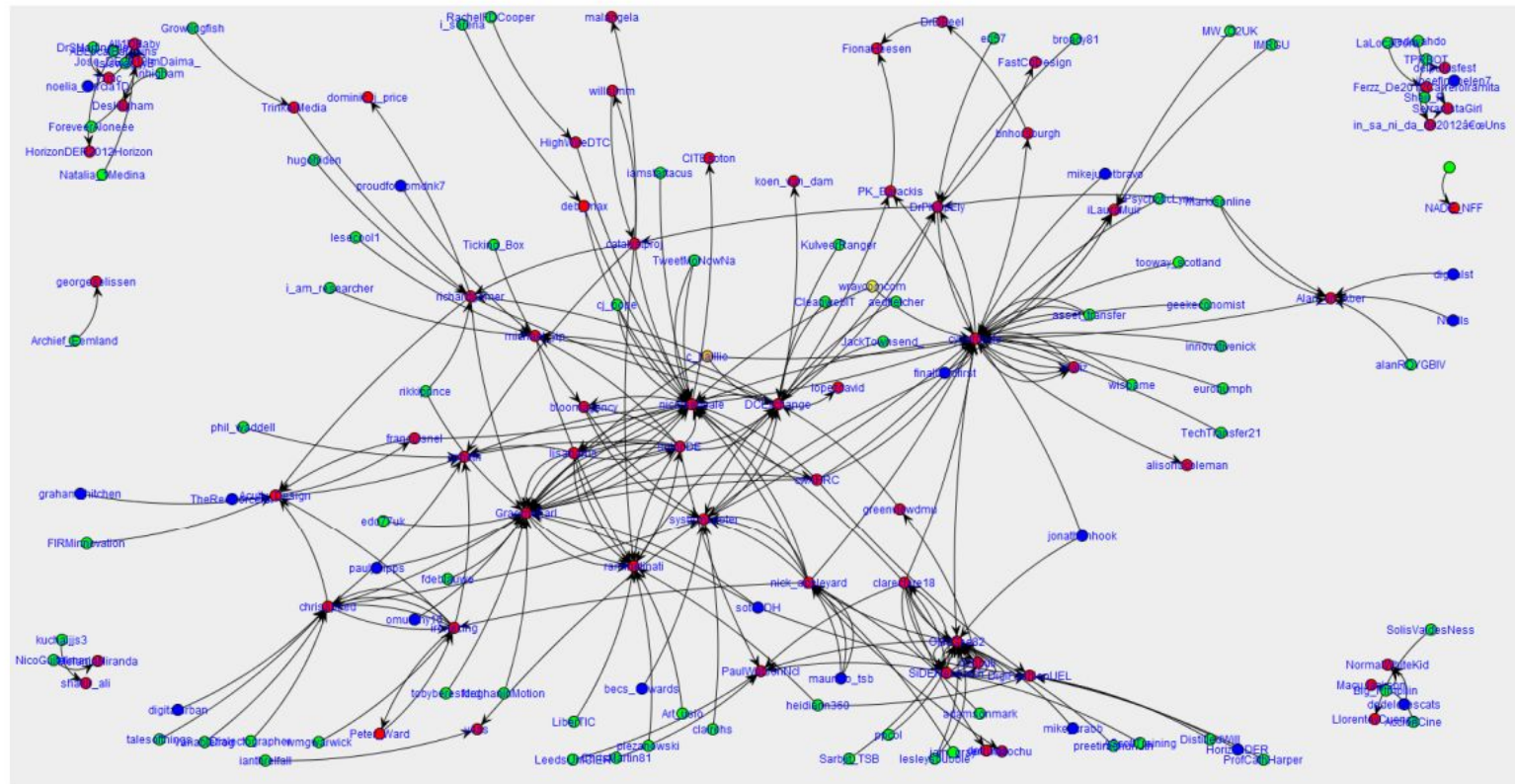
Applications of Network Analysis

Example: Food Chain



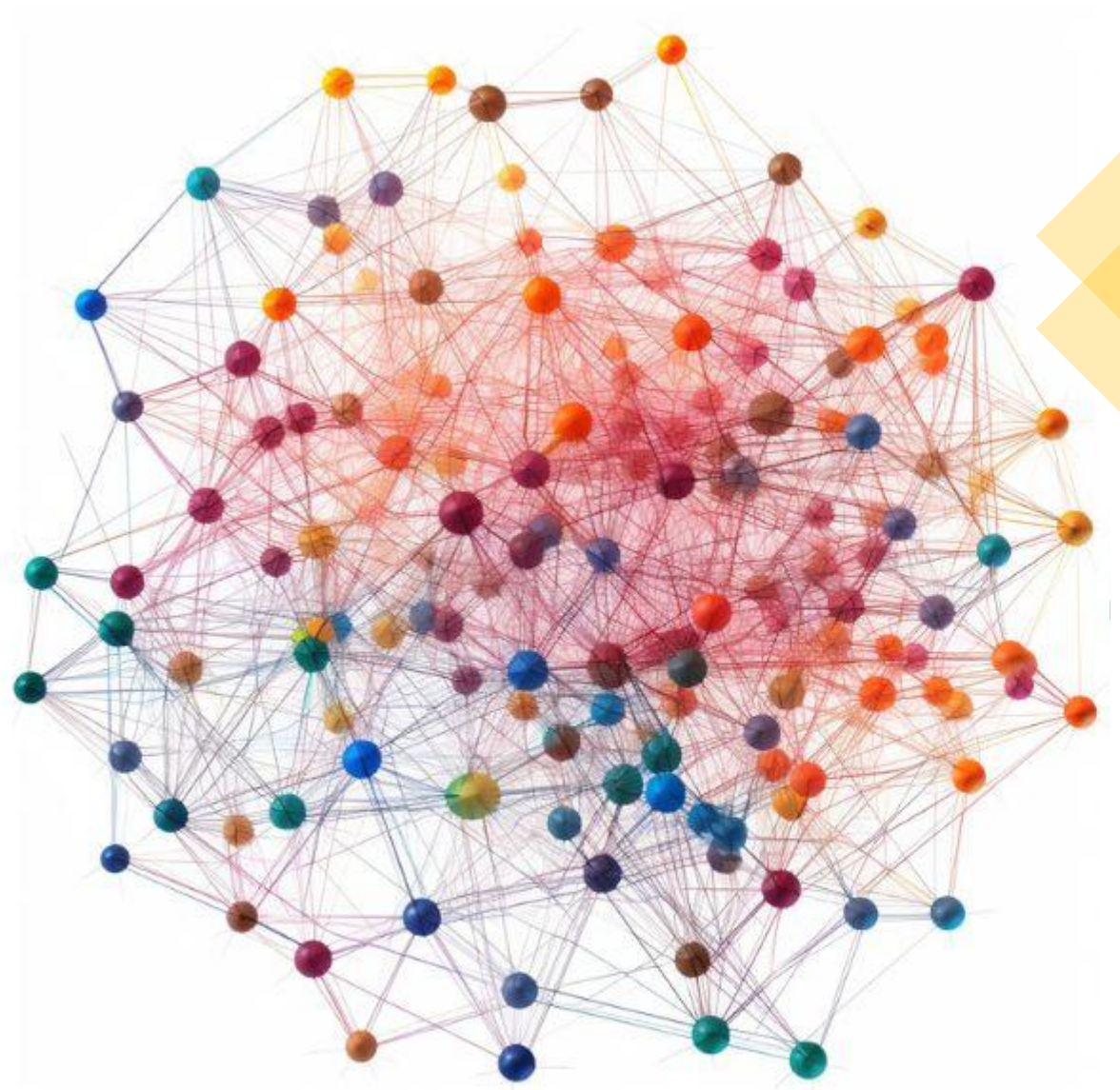
Applications of Network Analysis

Example: Retweets



Basic Structure

- ❖ In network analysis, the network structure refers to the **arrangement of and relations between the nodes and edges** in a network. It is a key aspect of network analysis as it can provide insights into the overall behavior of the network and its individual components.



Basic Structure

Here are some key elements of network structure:

- **Nodes:** the **individual entities** in the network. They can represent anything from people in a social network, to computers in a network infrastructure, to neurons in a neural network.
- **Edges:** represent the **relationships or connections between the nodes**. They can be directed (one-way relationship) or undirected (two-way relationship). The number of edges connected to the node is connected the degree of the node.
- **Network Topology:** refers to the **overall shape or structure of the network**. It can be regular (every node has the same number of connections), random (edges between nodes are formed randomly), scale-free (some nodes have many more connections than others), among others.
- **Clusters or Communities:** these groups of nodes are more **densely connected** with each other than any other node in the network. Identifying these can be important for understanding the structure and behavior of the network.
- **Centrality Measures:** metrics used to **identify the most important nodes in the network**. They can be based on the degree of the node, its closeness to other nodes, or its role in connecting different parts of the network.

Network Analysis Techniques

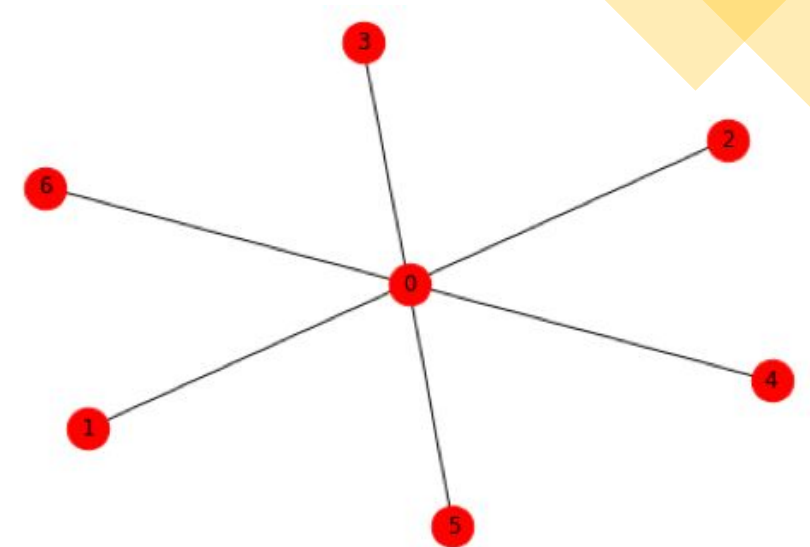
The following are the different types of network analysis techniques:

- **Centrality Measures:** used to **identify the most important nodes in a network**. There are several types of centrality measures, including degree centrality, betweenness centrality, and eigenvector centrality (a measure of a node's influence based on the number and quality of its connections). For example, in a social network, a person with **high degree centrality might be someone with many friends**, while a person with **high betweenness centrality might be someone who connects disparate groups** of people.
- **Community Detection:** used to **identify clusters or communities of nodes**. These are groups of nodes that are more **densely connected with each other** than to the rest of the network. In a social network, these could represent groups of friends or colleagues. There are several algorithms for community detection, such as the Louvain method and the Girvan-Newman method.
- **Network Modeling:** used to **model how networks form and grow**. For example, the Barabási–Albert model is a popular model for generating random scale-free networks, which are networks where some nodes have many more connections than others (like many real-world networks).
- **Network Dynamics:** used to study **how networks change over time**. This could involve studying how nodes and connections are added or removed, how communities form and dissolve, or how centrality measures change over time.

Centrality (Metrics)

A crucial application of network analysis is identifying the important node in a network. This task is called Measuring **Network Centrality**. In social network analysis, it can refer to the task of identifying the most influential member, or the representative of the group.

- **Degree centrality: number of edges a node has.** A node with a higher degree has higher centrality.
- **Eigenvector centrality:** adding to the degree of one node, the centralities of neighbor nodes are considered. As a result, the **eigenvector corresponding to the highest eigenvalue** of the adjacency matrix represents the centrality of nodes in the network.
- **Betweenness centrality: finds shortest path.** The **number of paths between two nodes** that go through the i -th node is considered as the i -th node's betweenness centrality.
- **Closeness centrality:** finds the **nodes with the slightest distance between them**. The **length of the path** from the i -th node to other nodes in the network is considered as the i -th node's closeness centrality.



Centrality

```
import networkx as nx
import numpy as np
import matplotlib.pyplot as plt

G = nx.Graph()
G.add_nodes_from(["A","B","C","D","E","F","G","H","I","J","K"])
G.add_edges_from([("A","C"),("B","C"),("C","D"),("D","E"),
                  ("D","G"),("A","G"),("F","H"),("G","H"),("H","I"),
                  ("I","J"),("I","K")])

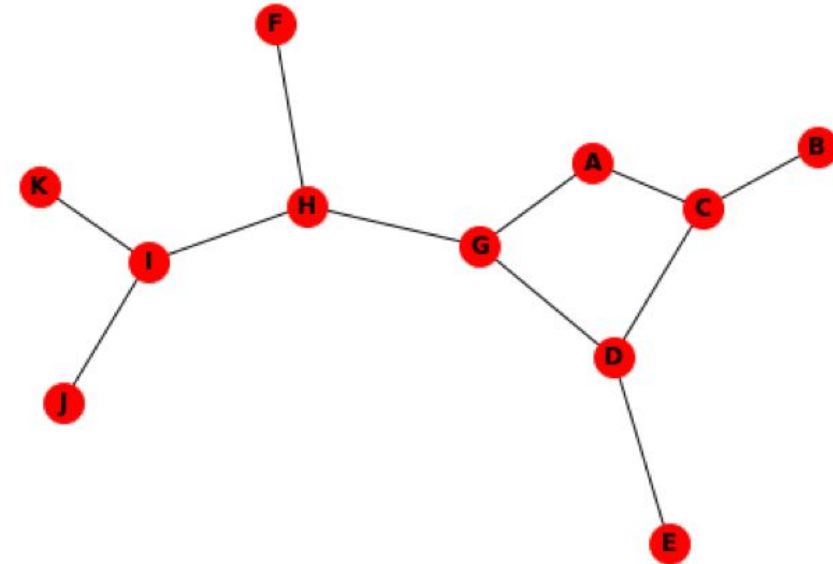
nx.draw(G, node_size=400, node_color='red', with_labels=True,
        font_weight='bold')

print("degree centrality:")
for k, v in sorted(nx.degree_centrality(G).items(), key=lambda x: -x[1]):
    print(str(k)+":"+ "{:.3}".format(v)+" ", end="")
print("\n")

print("eigenvector centrality:")
for k, v in sorted(nx.eigenvector_centrality(G).items(), key=lambda x: -x[1]):
    print(str(k)+":"+ "{:.3}".format(v)+" ", end="")
print("\n")

print("between centrality:")
for k, v in sorted(nx.betweenness_centrality(G).items(), key=lambda x: -x[1]):
    print(str(k)+":"+ "{:.3}".format(v)+" ", end="")
print("\n")

print("closeness centrality:")
for k, v in sorted(nx.closeness_centrality(G).items(), key=lambda x: -x[1]):
    print(str(k)+":"+ "{:.3}".format(v)+" ", end="")
print("\n")
```



```
degree centrality:
C:0.3 D:0.3 G:0.3 H:0.3 I:0.3 A:0.2 B:0.1 E:0.1 F:0.1 J:0.1 K:0.1

eigenvector centrality:
G:0.481 D:0.443 C:0.405 A:0.367 H:0.352 I:0.222 E:0.184 B:0.168 F:0.146 J:0.0918 K:0.0918

between centrality:
H:0.6 G:0.578 I:0.378 D:0.333 C:0.222 A:0.133 B:0.0 E:0.0 F:0.0 J:0.0 K:0.0

closeness centrality:
G:0.5 H:0.476 D:0.435 A:0.4 I:0.385 C:0.357 F:0.333 E:0.312 J:0.286 K:0.286 B:0.27
```

Network Analysis in Python

```
import networkx as nx
import matplotlib.pyplot as plt

# Create a new undirected graph
G = nx.Graph()

# Add nodes
G.add_nodes_from([1, 2, 3, 4, 5])

# Add edges
G.add_edges_from([(1, 2), (1, 3), (2, 3), (3, 4), (4, 5), (2, 5)])

# Draw the graph
nx.draw(G, with_labels=True, node_color='lightblue', node_size=1000, font_size=12, font_weight='bold')
plt.title('Network Analysis Diagram')
plt.show()

# Calculate and print performance metrics
print("Performance Metrics:")
print("- Number of nodes:", G.number_of_nodes())
print("- Number of edges:", G.number_of_edges())
print("- Average clustering coefficient:", nx.average_clustering(G))
print("- Diameter (longest shortest path):", nx.diameter(G))
print("- Degree centrality:")
for node, centrality in nx.degree_centrality(G).items():
    print(f"Node {node}: {centrality:.3f}")
```

Network Analysis in Python

```
import networkx as nx
import matplotlib.pyplot as plt

# Create a new undirected graph
G = nx.Graph()

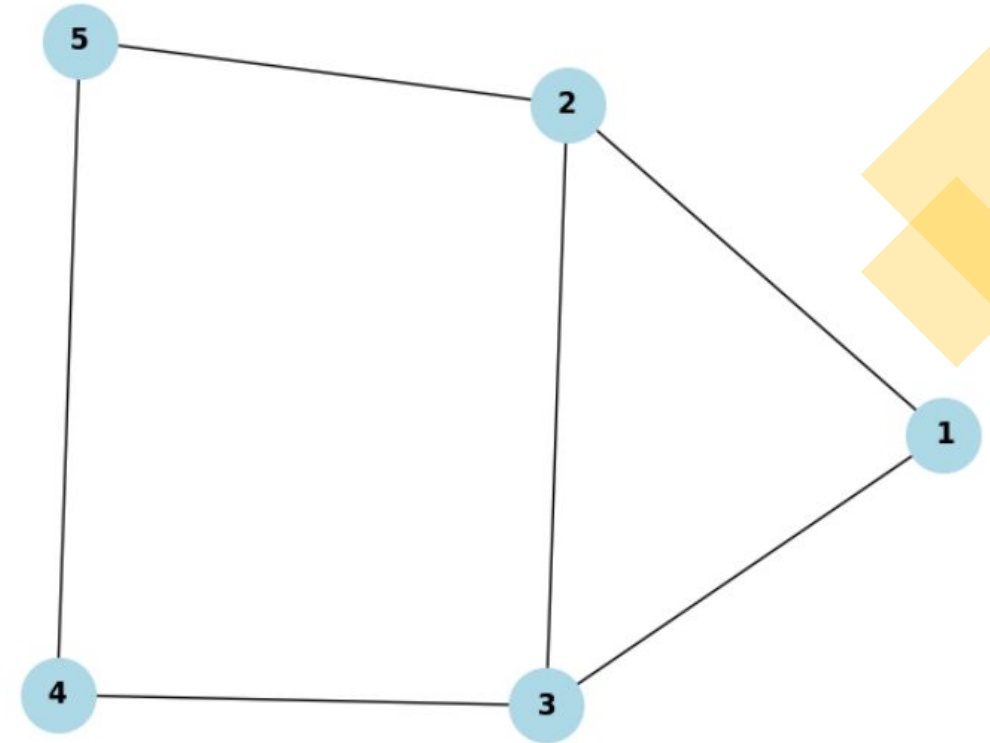
# Add nodes
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print("- Degree centrality:")
for node, centrality in nx.degree_centrality(G).items():
    print(f"Node {node}: {centrality:.3f}")
```

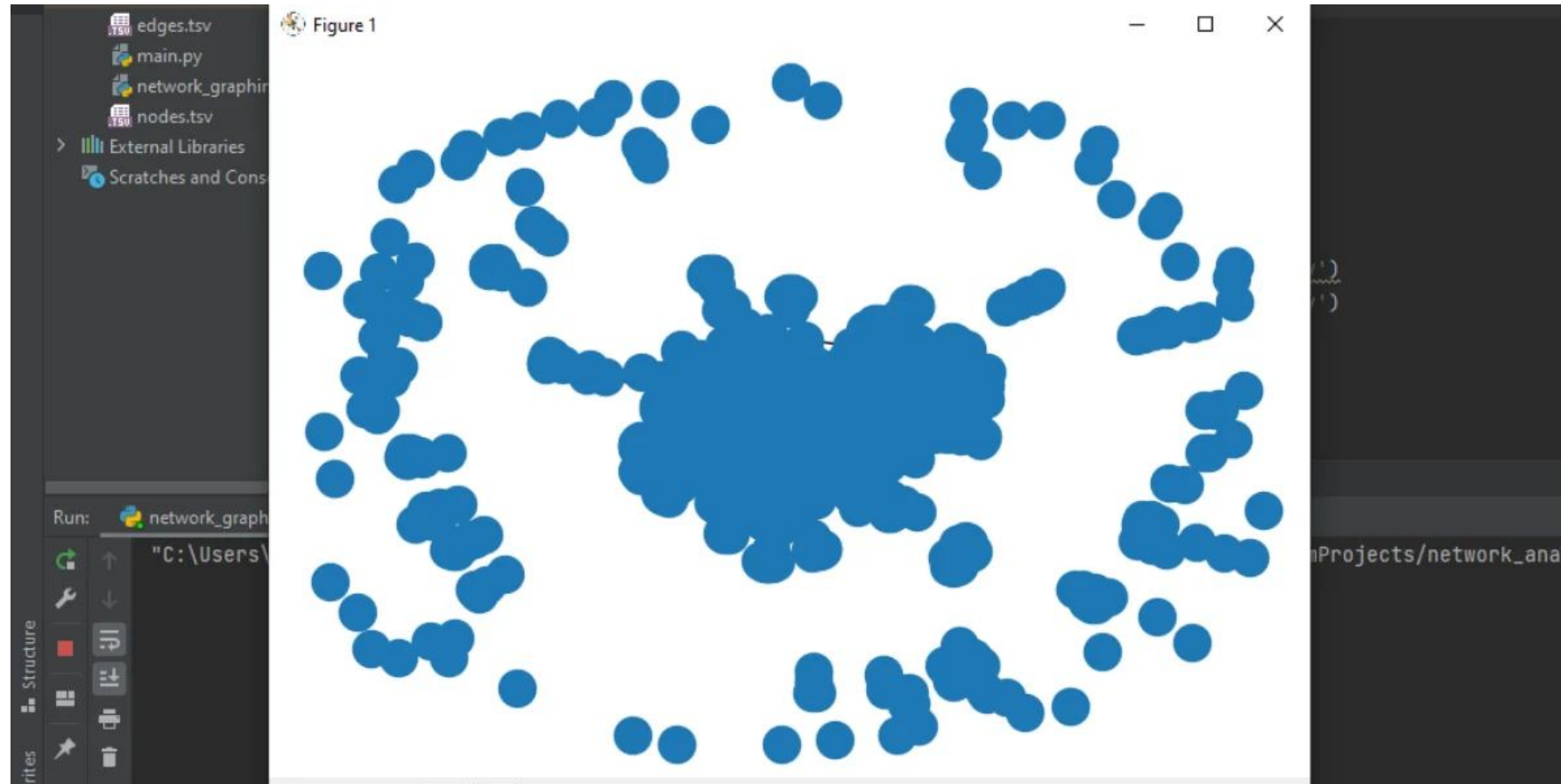
Network Analysis Diagram



Performance Metrics:

- Number of nodes: 5
- Number of edges: 6
- Average clustering coefficient: 0.3333333333333333
- Diameter (longest shortest path): 2
- Degree centrality:
 - Node 1: 0.500
 - Node 2: 0.750
 - Node 3: 0.750
 - Node 4: 0.500
 - Node 5: 0.500

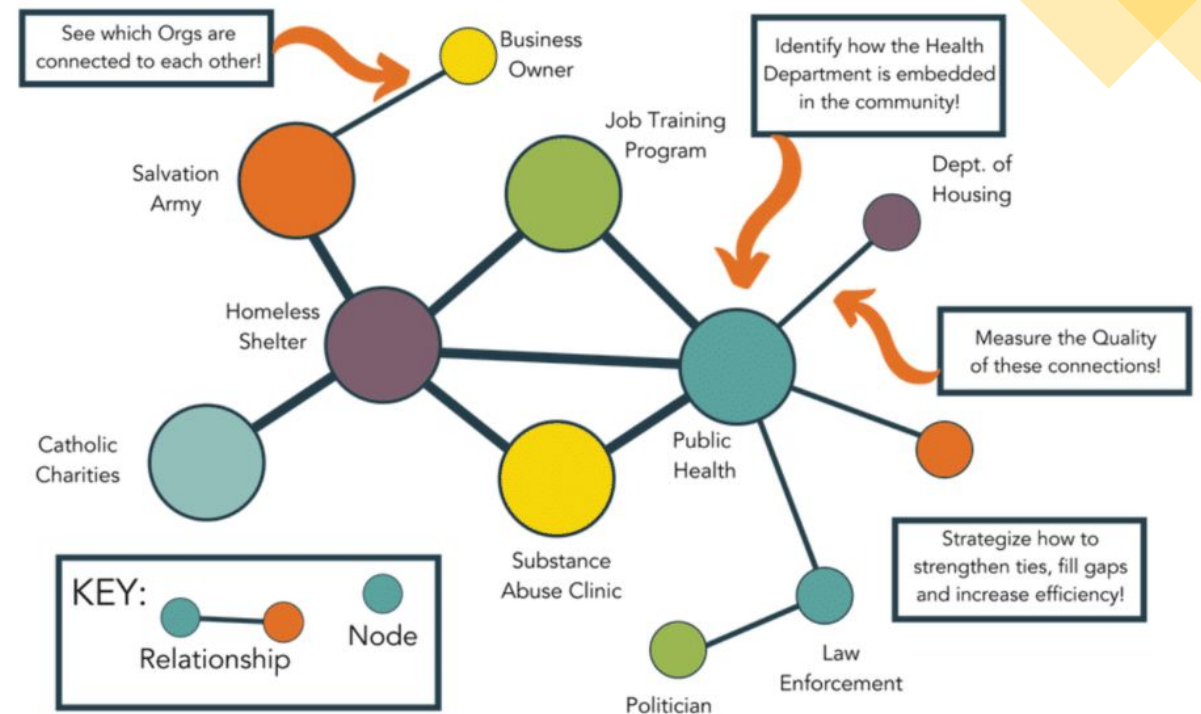
Network Analysis in Python



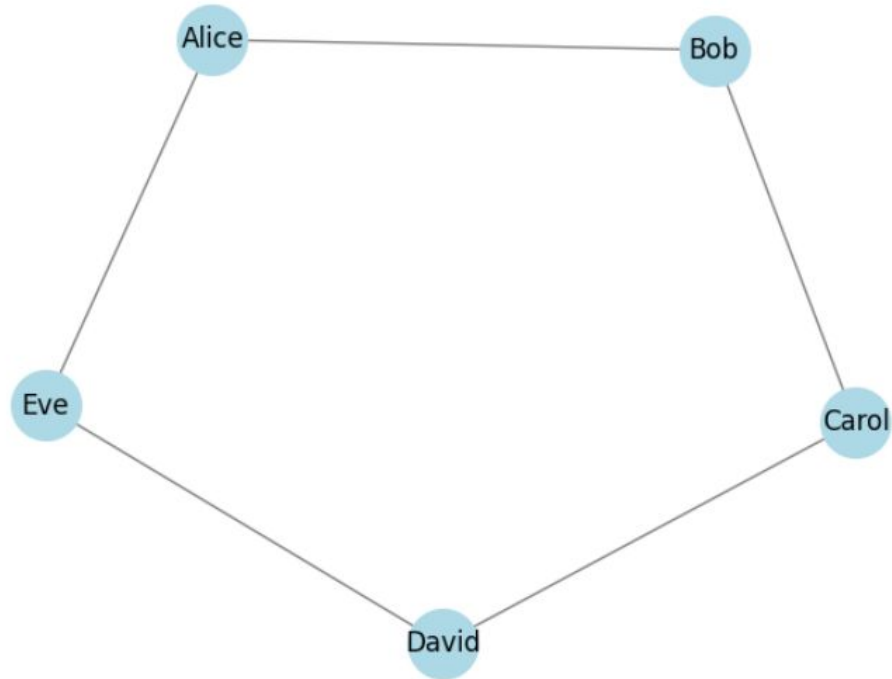
Python Code: <https://www.askpython.com/python/examples/network-analysis-in-python>

Social Network Analysis

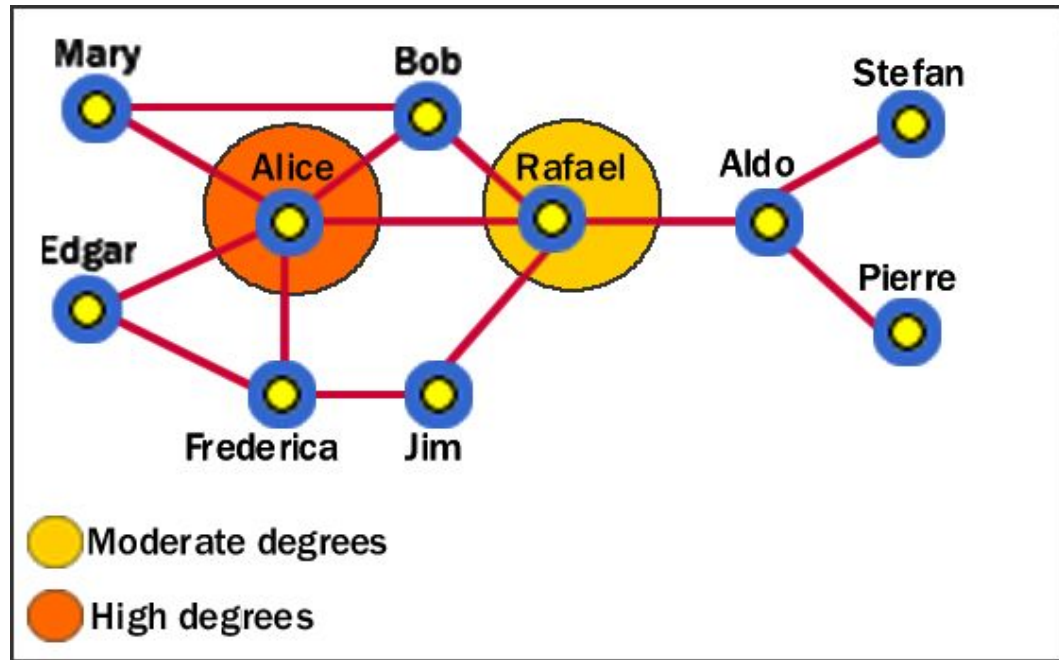
- ❖ SNA, is a research method used to visualize and **analyze relationships and connections between entities or individuals** within a network.
- ❖ In SNA, nodes represent individuals or entities while edges symbolize the relationships between them. For example, in an inter-organizational network, nodes might be companies, and edges could represent communication, collaboration, or competition.



Social Network Analysis



Number of nodes: 5
Number of edges: 5
Node degrees: {'Alice': 2, 'Bob': 2, 'Carol': 2, 'David': 2, 'Eve': 2}
Degree Centrality:
Alice: 0.500
Bob: 0.500
Carol: 0.500
David: 0.500
Eve: 0.500
Betweenness Centrality:
Alice: 0.167
Bob: 0.167
Carol: 0.167
David: 0.167
Eve: 0.167



References

<https://towardsdatascience.com/network-analysis-d734cd7270f8>

<https://web.stanford.edu/class/cs102/lectureslides/NetworksSlides.pdf>

<https://www.shiksha.com/online-courses/articles/understanding-network-analysis/>

<https://visiblenetworklabs.com/guides/social-network-analysis-101/>